

A disproof of conjecture 00000001323: the period set in $\text{Aut}(\mathbb{C}^2)$ is not $\{1, 2, 3, 4, 6\}$

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Abstract

Conjecture 00000001323 states that in the automorphism group $\text{Aut}(\mathbb{C}^2)$ the set of possible periods of elements whose Jacobian is (identically) the constant 1 is $\{1, 2, 3, 4, 6\}$. We show that the conjecture is false. For every positive integer m the diagonal linear map $A_m = \text{diag}(\zeta_m, \zeta_m^{-1})$, where ζ_m is a primitive m -th root of unity, is an element of $\text{Aut}(\mathbb{C}^2)$ with $\det A_m = 1$ and with order exactly m . Hence every m , and in particular $m = 5$, occurs as a period, while $5 \notin \{1, 2, 3, 4, 6\}$. We also explain the reading under which the statement would have been true (integer-coefficient automorphisms), and record a formal verification of the underlying finite-field witness in Lean 4.

1 The conjecture and the counterexample

Throughout, an element of $\text{Aut}(\mathbb{C}^2)$ means a polynomial automorphism of $\mathbb{C}^2$; its *period* (order) is the least positive integer n with $g^{\circ n} = \text{id}$, whenever it exists. The Jacobian of a polynomial map $F = (F_1, F_2)$ is the matrix-valued function

$$\text{Jac } F = \begin{pmatrix} \partial F_1 / \partial x_1 & \partial F_1 / \partial x_2 \\ \partial F_2 / \partial x_1 & \partial F_2 / \partial x_2 \end{pmatrix}.$$

Conjecture 00000001323 asserts:

In $\text{Aut}(\mathbb{C}^2)$, the set of possible periods of elements with $\det \text{Jac } F \equiv 1$ is $\{1, 2, 3, 4, 6\}$.

For $m \geq 1$ let $\zeta_m = \exp(2\pi i/m)$ be a primitive m -th root of unity and consider the linear map

$$A_m = \text{diag}(\zeta_m, \zeta_m^{-1}) = \begin{pmatrix} \zeta_m & 0 \\ 0 & \zeta_m^{-1} \end{pmatrix}. \tag{1}$$

Since A_m is linear, it is given by polynomial coordinate functions $A_m(x_1, x_2) = (\zeta_m x_1, \zeta_m^{-1} x_2)$, and it is invertible with inverse $\text{diag}(\zeta_m^{-1}, \zeta_m)$, so $A_m \in \text{GL}_2(\mathbb{C}) \subseteq \text{Aut}(\mathbb{C}^2)$.

Proposition 1. *For every $m \geq 1$ the map A_m of (1) satisfies*

$$\det \text{Jac } A_m \equiv 1 \quad \text{and} \quad \text{ord}(A_m) = m.$$

Proof. The Jacobian of a linear map is the constant matrix of that linear map, so $\text{Jac } A_m$ is the constant matrix A_m and

$$\det \text{Jac } A_m = \det A_m = \zeta_m \cdot \zeta_m^{-1} = 1,$$

which is identically 1 as a function on $\mathbb{C}^2$. For the order, compute powers using that a diagonal matrix is raised to powers entrywise:

$$(A_m)^k = \text{diag}(\zeta_m^k, \zeta_m^{-k}) = \text{diag}(e^{2\pi i k/m}, e^{-2\pi i k/m}).$$

This equals the identity $I = \text{diag}(1, 1)$ if and only if $\zeta_m^k = 1$, i.e. if and only if $m \mid k$ (because ζ_m is primitive of order m). The least positive such k is $k = m$. Hence $\text{ord}(A_m) = m$. $\square$

Corollary 2 (Conjecture 00000001323 is false). *Every positive integer occurs as the period of an element of $\text{Aut}(\mathbb{C}^2)$ whose Jacobian is identically 1. Consequently the period set is all of $\mathbb{N}_{>0}$, which strictly contains $\{1, 2, 3, 4, 6\}$.*

Proof. Immediate from Proposition 1, taking m arbitrary. Already $m = 5$ suffices: $\text{ord}(A_5) = 5$ and $\det \text{Jac } A_5 \equiv 1$, but $5 \notin \{1, 2, 3, 4, 6\}$ (indeed $5 \neq 1, 2, 3, 4, 6$). $\square$

The counterexample is as concrete as possible: for $m = 5$,

$$A_5 = \text{diag}(\zeta_5, \zeta_5^{-1}), \quad \zeta_5 = e^{2\pi i/5}, \quad \det A_5 = 1, \quad \text{ord}(A_5) = 5.$$

2 Non-linearity does not help

One might try to rescue the conjecture by insisting on *non-linear* automorphisms, since the diagonal witnesses above are linear. This also fails.

Proposition 3. *For every $m \geq 1$ there exists a non-linear polynomial automorphism $F \in \text{Aut}(\mathbb{C}^2)$ with $\det \text{Jac } F \equiv 1$ and $\text{ord}(F) = m$.*

Proof. Let $H \in \text{Aut}(\mathbb{C}^2)$ be any non-linear polynomial automorphism, for instance $H(x_1, x_2) = (x_1 + x_2^2, x_2)$, and put $F = H \circ A_m \circ H^{-1}$, where A_m is the linear witness (1). Then F is a polynomial automorphism, it is non-linear (conjugation preserves linearity, and F is linear only if A_m is), and it has the same order as A_m , namely m , since conjugation is an isomorphism of groups. By the chain rule,

$$\text{Jac } F = (\text{Jac } H) \circ H^{-1} \cdot (\text{Jac } A_m) \circ H^{-1} \cdot \text{Jac}(H^{-1}),$$

and at each point u with $v = H^{-1}(u)$ the right-hand side equals $\text{Jac } H(v) \cdot A_m \cdot (\text{Jac } H(v))^{-1}$. Its determinant is

$$\det \text{Jac } F(u) = \det A_m = 1,$$

using multiplicativity of the determinant and $\det(\text{Jac } H(v))^{-1} = (\det \text{Jac } H(v))^{-1}$; in particular $\det \text{Jac } F$ is the constant function 1 (it is independent of u). Thus F is a non-linear counterexample of period m . $\square$

Remark 4 (Conjugating by a non-linear map). The construction even yields a polynomial automorphism of *degree* $2 \deg H$ exhibiting any prescribed period, so non-linearity imposes no restriction on the achievable periods in $\text{Aut}(\mathbb{C}^2)$.

3 Why the claim *would* be true with integer coefficients

We record honestly the reading under which the conjectured set $\{1, 2, 3, 4, 6\}$ is correct, so that the failure above is not misread as a failure of the well-known restriction it resembles.

If one restricts to automorphisms with integer (or, equivalently for this purpose, rational) coefficients, the situation is different. The linear part then lies in $\text{GL}_2(\mathbb{Z})$, and an element of finite order in $\text{GL}_2(\mathbb{Z})$ with determinant 1 lies in the finite group $\text{SL}_2(\mathbb{Z})$; the finite-order elements of $\text{SL}_2(\mathbb{Z})$ have orders exactly

$$\{1, 2, 3, 4, 6\},$$

realised by rotations of orders 1, 2, 3, 4, 6 (equivalently by the cyclotomic polynomials Φ_n with $\varphi(n) \leq 2$, namely $n = 1, 2, 3, 4, 6$). This number-theoretic restriction comes from the rational/integral structure; it is *not* a restriction valid over $\mathbb{C}$.

The statement of conjecture 00000001323, however, is explicitly about $\text{Aut}(\mathbb{C}^2)$ over the complex numbers, with no integrality hypothesis on the coefficients. The witness (1) uses the coefficient $\zeta_5 \in \mathbb{C} \setminus \mathbb{Z}$, so it lies in the class the conjecture quantifies over, and it has period 5. Hence the conjecture, as stated, is false.

Remark 5 (On the phrase “Jacobian constant 1”). The phrase could in principle be read in a weaker or a stronger sense:

- “ $\det \text{Jac } F$ is the constant function 1” (the natural reading, and the one used above). Under this reading every m occurs, by Proposition 1.
- “ $\text{Jac } F$ is literally the identity matrix”. Under this reading the period set would be contained in $\{1\}$, since $\text{Jac } F = I$ forces F to be a translation, and a nontrivial translation has infinite order; so the only finite period would be 1. This is even further from $\{1, 2, 3, 4, 6\}$.

Either way the conjecture fails.

4 Machine verification

The algebraic core of the refutation is the order and determinant computation for A_m ; it is coefficient-free and holds over any commutative ring in which a primitive m -th root of unity and its inverse exist. The accompanying Lean 4 development (`lean4/`) formalises the algebraically identical finite-field model over $\mathbb{F}_{11}$: the element $3 \in \mathbb{F}_{11}$ has multiplicative order exactly 5 (as $5 \mid 10 = |\mathbb{F}_{11}^\times|$), and the diagonal map $\text{diag}(3, 3^{-1}) = \text{diag}(3, 3^4)$ has determinant 1 and order exactly 5. This certifies the required determinant/order computation, which transfers verbatim to $\mathbb{C}$ with $\zeta_5 = e^{2\pi i/5}$ because no field-specific property beyond commutativity of multiplication is used.